Rule-Based Chatbot Report

Topic: College Help Desk

1. Introduction

This project is a rule-based chatbot built in Python that answers common student queries about a college: admission, courses, fees, scholarships, hostel, library, exams, results, placements, canteen, transport, sports and contact details. It also has a live web version with the same rules.

2. Working

User input is converted to lowercase and checked against keywords using an if-elif-else chain. The first matching condition returns its fixed answer. If no keyword matches, the else block returns a default help message. Typing 'bye' ends the conversation.

3. Tools Used

Python 3 (if-else conditions, functions, while loop, input/print). Web version: React/TypeScript, hosted on Lovable.

4. Sample Conversations

#	User Question	Chatbot Response
1	Hello	Hello! Welcome to the College Help Desk. How can I help you today?
2	What is the admission process?	Admissions are open from June to August. Apply online on the college portal with your 12th marksheet, ID proof and photographs.
3	Which courses are available?	We offer B.Tech (CSE, IT, ME, CE), BCA, BBA, B.Com, MCA and MBA programs.
4	What is the fee structure?	B.Tech: Rs. 1,20,000/year, BCA: Rs. 60,000/year, BBA: Rs. 55,000/year, MBA: Rs. 1,00,000/year. Fees can be paid in two installments.
5	Is there any scholarship?	Merit scholarships are given to students scoring above 85%. Government scholarships are also available via the state portal.
6	Tell me about hostel facility	Separate hostels for boys and girls are available with Wi-Fi, mess and 24x7 security. Hostel fee is Rs. 70,000/year including food.
7	What are library timings?	The library is open 8:00 AM to 8:00 PM, Monday to Saturday, with 50,000+ books.
8	When are the exams?	Mid-semester exams are held in October/March and end-semester exams in December/May. Check the portal for the exact timetable.
9	When will results come?	Results are published on the college web portal within 30 days after exams.
10	How are the placements?	Our placement cell has 150+ recruiting companies. Highest package last year was 18 LPA and average was 4.5 LPA.
11	Do you have bus transport?	College buses run on 12 routes across the city. Bus fee is Rs. 15,000/year.
#	User Question	Chatbot Response

12	What sports facilities are there?	We have cricket, football, basketball, volleyball courts and an indoor gym.
13	Give me contact details	Phone: +91 98765 43210 | Email: info@college.edu.in
14	Where is the college located?	The college is located at University Road, Ahmedabad, Gujarat.
15	What is the weather today?	Sorry, I didn't understand that. You can ask about admission, courses, fees, hostel, library, exams, results, placement, transport, sports or contact.
16	Thank you	You're welcome! Happy to help.
17	bye	Goodbye! Have a great day.

5. Code Snippet

```
def get_response(user_input):
    msg = user_input.lower().strip()
    if "admission" in msg:
        return "Admissions are open from June to August..."
    elif "fee" in msg:
        return "B.Tech: Rs. 1,20,000/year ..."
    ...
    else:
        return "Sorry, I didn't understand that..
```

6. Conclusion

The chatbot shows how simple if-else rules can automate answers to frequent questions. It can be improved with more rules, NLP, or a database of answers.